

Outline

- 1) Maximum Likelihood Estimator
- 2) Asymptotic Distribution of MLE
- 3) Consistency of MLE

Asymptotic Dist. of MLE

Under mild conditions, $\hat{\theta}_{MLE}$ is asy. Gaussian, efficient

We will be interested in $\ell_n(\theta; X)$ as a function of θ

Notate "true" value as θ_0 ($X \sim P_{\theta_0}$)

Derivatives of ℓ_n at θ_0 : ($\theta_0 \in \Theta^0$)

$$\nabla \ell_1(\theta_0; X_i) \stackrel{iid}{\sim} (0, J_1(\theta_0))$$

$$\frac{1}{\sqrt{n}} \nabla \ell_n(\theta_0; X) = \sqrt{n} \cdot \frac{1}{n} \sum \nabla \ell_1(\theta_0; X_i) \xrightarrow{P_{\theta_0}} N(0, J_1(\theta_0))$$

$$\frac{1}{n} \nabla^2 \ell_n(\theta_0; X) \xrightarrow{P_{\theta_0}} \mathbb{E}_{\theta_0} \nabla^2 \ell_1(\theta_0; X_i) = -J_1(\theta_0)$$

Proof sketch:

$$0 = \nabla \ell_n(\hat{\theta}_n; X) = \nabla \ell_n(\theta_0) + \nabla^2 \ell_n(\tilde{\theta}_n) (\hat{\theta}_n - \theta_0)$$

↙ between $\theta_0, \tilde{\theta}_n$

$$\sqrt{n} (\hat{\theta}_n - \theta_0) = - \underbrace{\left(\frac{1}{n} \nabla^2 \ell_n(\tilde{\theta}_n) \right)^{-1}}_{\text{(want)} \xrightarrow{P} J(\theta_0)^{-1}} \underbrace{\frac{1}{\sqrt{n}} \nabla \ell_n(\theta_0)}_{\Rightarrow N_d(0, J(\theta_0))}$$

$$\Rightarrow N_d(0, J(\theta_0)^{-1})$$

More rigorous proof later, but note
we need consistency of $\hat{\theta}_n$ first to even
justify Taylor expansion

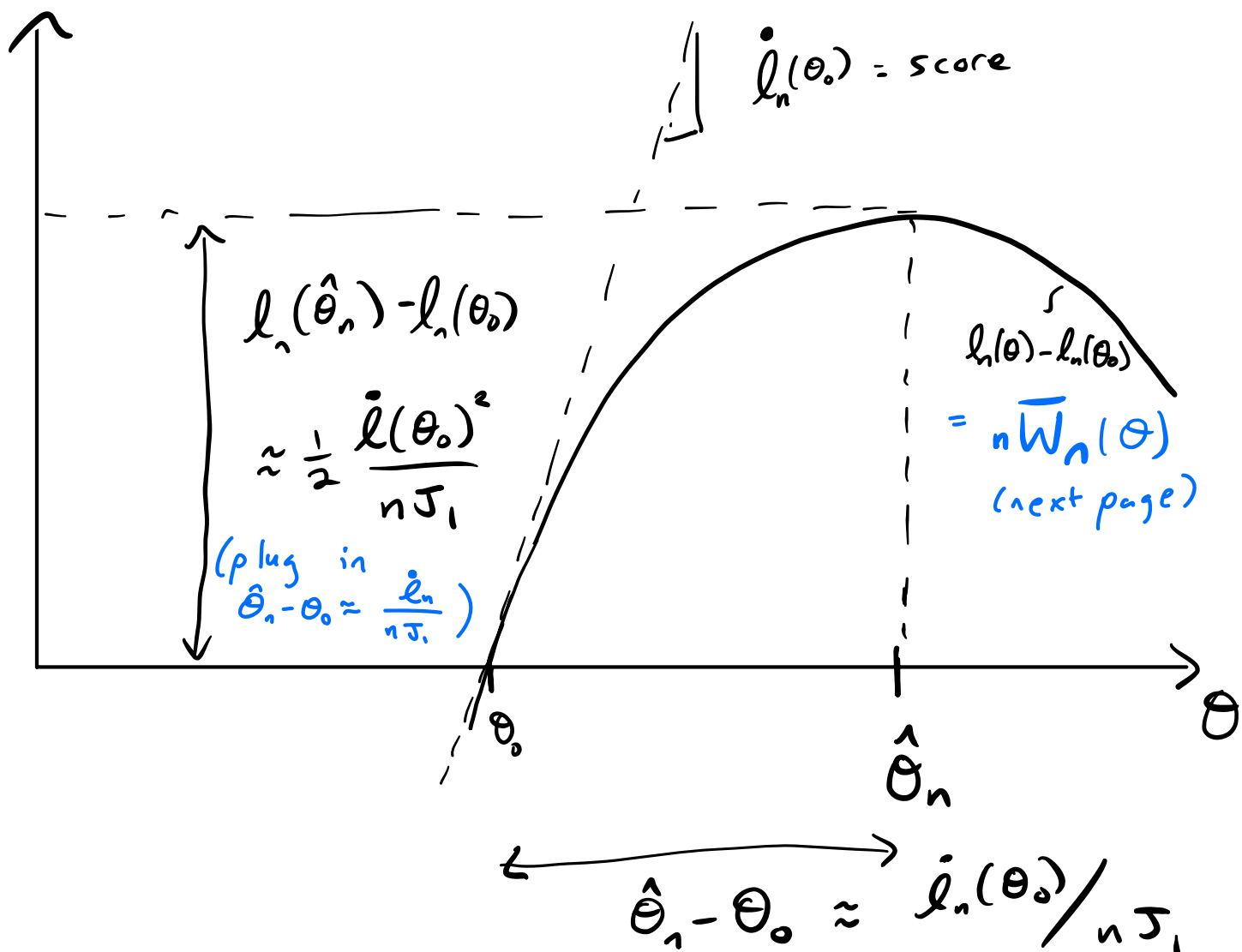
Asymptotic Picture ($d=1$)

Recall $(\ell_n(\theta) - \ell_n(\theta_0))_{\theta \in \mathcal{U}}$ is minimal suff.

Quadratic approximation near θ_0 :

$$\ell_n(\theta) - \ell_n(\theta_0) \approx \underbrace{\dot{\ell}_n(\theta_0)}_{\approx N(0, nJ_1(\theta_0))} (\theta - \theta_0) + \frac{1}{2} \underbrace{\ddot{\ell}_n(\theta_0)}_{\approx -nJ_1(\theta_0)} (\theta - \theta_0)^2$$

Gaussian linear term Deterministic curvature



Consistency of MLE

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_{\theta_0}, \quad \hat{\theta}_n \in \arg \max_{\theta \in \Theta} \ell_n(\theta; X)$$

[Will be ok if $\hat{\theta}_n$ comes close to maximizing ℓ_n]

Question: When does $\hat{\theta}_n \xrightarrow{P} \theta_0$?

Assume model identifiable ($P_\theta \neq P_{\theta_0}$ for $\theta \neq \theta_0$)

Recall KL Divergence:

$$D_{KL}(\theta_0 \parallel \theta) = \mathbb{E}_{\theta_0} \log \frac{P_{\theta_0}(X_i)}{P_\theta(X_i)}$$

$$-D_{KL}(\theta_0 \parallel \theta) \leq \log \mathbb{E}_{\theta_0} \frac{P_\theta(X_i)}{P_{\theta_0}(X_i)} \quad \leftarrow \text{(note switch)}$$

$$= \log \int \frac{P_\theta(x)}{P_{\theta_0}(x)} \cancel{P_{\theta_0}(x)} d\mu(x)$$

$x: P_{\theta_0}(x) > 0$

$$\leq \log 1 = 0$$

(Jensen)

strict ineq unless $\frac{P_\theta}{P_{\theta_0}}$ const. (i.e., unless $P_\theta = P_{\theta_0}$)

Let $W_i(\theta) = \ell_i(\theta; X_i) - \ell(\theta_0; X_i)$, $\bar{W}_n = \frac{1}{n} \sum W_i$

Note $\hat{\theta}_n \in \arg \max_{\theta \in \Theta} \bar{W}_n(\theta)$ too

$$\begin{aligned}\bar{W}_n(\theta) &\xrightarrow{P} \mathbb{E}_{\theta_0} W_i(\theta) \\ &= -D_{KL}(\theta_0 \parallel \theta) \\ &\leq 0, \text{ equality iff } \theta = \theta_0\end{aligned}$$

But not enough:

- MLE $\hat{\theta}_n$ depends on entire function $\bar{W}_n(\cdot)$
- need uniform convergence in θ

Def For compact K let $C(K) = \{f: K \rightarrow \mathbb{R}, \text{cts}\}$

For $f \in C(K)$ let $\|f\|_\infty = \sup_{t \in K} |f(t)|$

$f_n \xrightarrow{(\xrightarrow{P})} f$ in this norm if $\|f_n - f\|_\infty \xrightarrow{(\xrightarrow{P})} 0$

Thm (LLN for random functions)

Assume K compact, $W_1, W_2, \dots \in C(K)$ iid.

$$\mathbb{E} \|W_i\|_\infty < \infty, \quad \mu(t) = \mathbb{E} W_i(t)$$

Then $\mu(t) \in C(K)$

$$\text{and } \mathbb{P}(\|\tfrac{1}{n} \sum W_i - \mu\|_\infty > \varepsilon) \rightarrow 0$$

(i.e., $\bar{W}_n \xrightarrow{P} \mu$ in $\|\cdot\|_\infty$, or $\|\bar{W}_n - \mu\|_\infty \xrightarrow{P} 0$)

Theorem (Consistency of MLE for compact Θ)

$X_1, \dots, X_n \stackrel{iid}{\sim} p_{\theta_0}$, $\mathcal{P}$ has densities p_{θ} , $\theta \in \Theta$

- Assume
- $\log p_{\theta}(x)$ cts in Θ , all $x \in \mathcal{X}$
 - Θ compact
 - $\mathbb{E}_{\theta_0} \left[\sup_{\theta \in \Theta} |W_i(\theta)| \right] < \infty$ = $\mathbb{E}_{\theta_0} \sup_{\theta} |\ell_i(\theta; X_i) - \ell_i(\theta_0; X_i)|$
 - Model identifiable

Then $\hat{\theta}_n \xrightarrow{P} \theta_0$ if $\hat{\theta}_n \in \arg\max \ell_n(\theta; X)$

Proof $W_i \in C(\Theta)$ iid, mean $\mu(\theta) = -D_{KL}(\theta_0 \parallel \theta)$
 $\mu(\theta_0) = 0$, $\mu(\theta) < 0 \quad \forall \theta \neq \theta_0$ ($\theta_0 = \arg\min \mu$)

By definition, $\hat{\theta}_n$ maximizes $\bar{w}_n$,

$$\mathcal{J}_n = \|\bar{w}_n - \mu\|_{\infty} \xrightarrow{P} 0.$$

Fix $\varepsilon > 0$, want to show $\mathbb{P}_{\theta_0}(\|\theta - \theta_0\| \geq \varepsilon) \rightarrow 0$

Let $\tilde{\Theta}_{\varepsilon} = \Theta \setminus B_{\varepsilon}(\theta_0) = \{\theta \in \Theta : \|\theta - \theta_0\| \geq \varepsilon\}$ (compact)

$$\text{Let } \mu_{\varepsilon}^* = \max_{\theta \in \tilde{\Theta}_{\varepsilon}} \mu(\theta) < 0 = \mu(\theta_0)$$

$$w_{\varepsilon}^* = \max_{\theta \in \tilde{\Theta}_{\varepsilon}} \bar{w}_n(\theta)$$

$$\begin{aligned} \mathbb{P}_{\theta_0}(\|\theta - \theta_0\| \geq \varepsilon) &\leq \mathbb{P}_{\theta_0}(\underbrace{w_\varepsilon^*}_{\leq \mu_\varepsilon^* + \delta_n} \geq \underbrace{\bar{w}_n(\theta_0)}_{\geq -\delta_n}) \\ &\leq \mathbb{P}_{\theta_0}(2\delta_n \geq \underbrace{-\mu_\varepsilon^*}_{> 0}) \rightarrow 0 \quad \square \end{aligned}$$

Note We usually care about non-compact parameter spaces, need some extra assumption to get us there.

Corollary Same assumptions except now $\Theta = \mathbb{R}^d$, ^(non-compact)
but there's some $R < \infty$ large enough so

$$\mathbb{P}_{\theta_0}(\|\hat{\theta}_n - \theta_0\| > R) \rightarrow 0$$

$$\text{Then } \hat{\theta}_n \xrightarrow{P} \theta_0$$

Proof Let $\tilde{\Theta} = \{\theta : \|\theta - \theta_0\| \leq R\}$, $\tilde{\theta}_n = \arg \max_{\theta \in \tilde{\Theta}} p_\theta(x)$

Then $\tilde{\theta}_n \rightarrow \theta_0$ by assumption

$$\mathbb{P}_{\theta_0}(\hat{\theta}_n \neq \tilde{\theta}_n) = \mathbb{P}_{\theta_0}(\hat{\theta}_n \notin \tilde{\Theta}) \rightarrow 0$$

$$\text{so } \hat{\theta}_n - \tilde{\theta}_n \xrightarrow{P} 0 \Rightarrow \hat{\theta}_n = \tilde{\theta}_n + (\hat{\theta}_n - \tilde{\theta}_n) \xrightarrow{P} \theta_0 \quad \square$$

So the only thing we actually need to worry about is if $\hat{\theta}_n$ is extremely far away from θ_0 with non-negligible Prob.

Theorem (Asymptotic distribution of MLE)

$X_1, \dots, X_n \stackrel{iid}{\sim} p_{\theta_0}$ $\mathcal{P}$ has densities p_{θ} , $\theta \in \Theta$

Assume

- $\mathcal{P}$ identifiable

- Θ compact

- $\mathbb{E}_{\theta_0} \left[\sup_{\theta \in \Theta} |W_i(\theta)| \right] < \infty$

- $l(\theta; x) = \log p_{\theta}(x)$ has two cts derivatives in θ

- $\mathbb{E}_{\theta_0} \sup_{\theta \in \Theta} \|\nabla^2 l_i(\theta; x_i)\| < \infty$

- $J_1(\theta_0) = \mathbb{E}_{\theta_0} \nabla^2 l_1(\theta_0; x_i) \succ 0$ ↙ positive definite

Then $\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, J_1(\theta_0)^{-1})$

Proof From before, we had

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \left(-\frac{1}{n} \nabla^2 \ell_n(\tilde{\theta}_n) \right)^{-1} \nabla \ell_n(\theta_0)$$

for $\tilde{\theta}_n$ between θ_0 and $\hat{\theta}_n$

Previous result shows $\hat{\theta}_n \xrightarrow{P} \theta_0$, so $\tilde{\theta}_n \xrightarrow{P} \theta_0$ also

Define $V_i(\theta) = -\nabla^2 l_i(\theta; x_i) \in C(\Theta)$, $\mathbb{E}_{\theta_0} \|V_i\|_{\infty} < \infty$
by assumption

Then $v(\theta) = \mathbb{E}_{\theta_0} V_i(\theta) \in C(\Theta)$, $v(\theta_0) = J_1(\theta_0)$

$$\bar{V}_n(\theta) = \frac{1}{n} \sum V_i(\theta), \quad \|\bar{V}_n - v\|_{\infty} \xrightarrow{P} 0$$

$$\begin{aligned}
\| -\frac{1}{n} \nabla^2 \ell_n(\tilde{\theta}_n) - J_1(\theta_0) \| &\leq \| \bar{V}_n(\tilde{\theta}_n) - v(\tilde{\theta}_n) \| + \| v(\tilde{\theta}_n) - v(\theta_0) \| \\
&\leq \underbrace{\| \bar{V}_n - v \|_\infty}_{\xrightarrow{P} 0} + \underbrace{\| v(\tilde{\theta}_n) - v(\theta_0) \|}_{\xrightarrow{P} 0 \text{ (cts mapping)}}
\end{aligned}$$

Hence $\left(-\frac{1}{n} \nabla^2 \ell_n(\tilde{\theta}_n) \right)^{-1} \xrightarrow{P} J_1(\theta_0)^{-1}$ (cts mapping)

And $\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N_d(0, J_1(\theta_0)^{-1})$ (Slutsky) $\square$

Note In this proof we played a bit fast and loose with our LLN for random functions, which we only stated for real-valued w_n

So technically we have only justified for $d=1$

But proof works for $d>1$